

LECTURE

Debugging Terraform



Why Would You Need Detailed Logs?



Investigating unexpected Terraform behavior

Preparing bug reports for HashiCorp or provider teams

Understanding exactly what Terraform is doing under the hood



The **TF_LOG** Environment Variable



Setting **TF_LOG** Enables Detailed Logging

```

$ export TF_LOG=TRACE      //Linux/MacOS/Bash
> $env:TF_LOG="TRACE"      //Windows Powershell

$ terraform plan
<detailed logs>

```

Most Verbose (Detailed)

Less Verbose



Separate Core and Provider Logging



```
TF_LOG_CORE=TRACE
```

- ✦ Logging can be enabled separately for Terraform Core itself
- ✦ Takes the same level arguments (trace, debug, info, etc.)
- ✦ Provide logs to the Terraform core team (support)

```
TF_LOG_PROVIDER=TRACE
```

- ✦ Enable logging for only the providers
- ✦ Takes the same level arguments (trace, debug, info, etc.)
- ✦ Offer logs to the provider developers to troubleshoot bugs

Saving Logs to a File



Persist Logs using the TF_LOG_PATH env

```

$ export TF_LOG=TRACE
$ export TF_LOG_PATH=terraform.log

```

TF_LOG (or TF_LOG_CORE/TF_LOG_PROVIDER) must also be set

Logs are appended to the file

File is created automatically if it doesn't exist